

G04 Gate A

Zero-to-Job-Ready Data Engineer 2026 | Protected assessment RC1

Protected Gate conditions

Learner book, review files, AI solving and tutorial/web solution hunting are closed. Use only the Gate brief, supplied Gate data and explicitly permitted command/reference help.

Scenario

TransitFlow: daily trip events arrive as CSV/API-like JSON and must become a trusted PostgreSQL serving table coordinated as a recoverable multi-step pipeline.

Required build / evidence

1. Declare source/landing/staging/serving grains and preserve raw evidence for every input.
2. Validate CSV/JSON contracts, quarantine defects and reconcile source rows to staged/quarantined outcomes.
3. Load governed PostgreSQL tables with keys/constraints and rerun-safe upsert behavior.
4. Apply deterministic transformations/deduplication and protect key/relationship/business invariants with tests.
5. Implement incremental state with a tie-breaker and prove state advances only after publish + reconciliation success.
6. Containerize or document the Docker execution boundary, and create an Airflow-style DAG/task design for ingest -> validate -> transform -> publish -> reconcile.
7. Inject one transient failure and one data-quality failure; show retry policy, no false state progress and safe rerun/recovery.
8. Produce run metadata, lineage, SQL validation and a handoff/runbook.

Pass standard

No unresolved critical failure. Correctness, integrity, reliability and reproducibility must be demonstrated with saved evidence; a high average cannot hide silent wrong data or secret exposure.